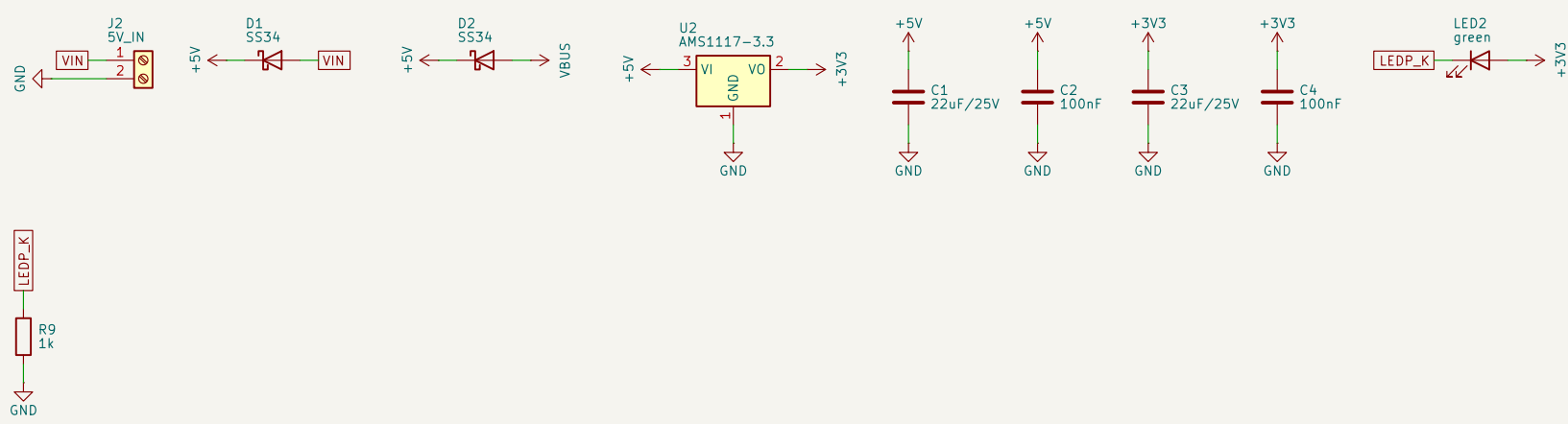
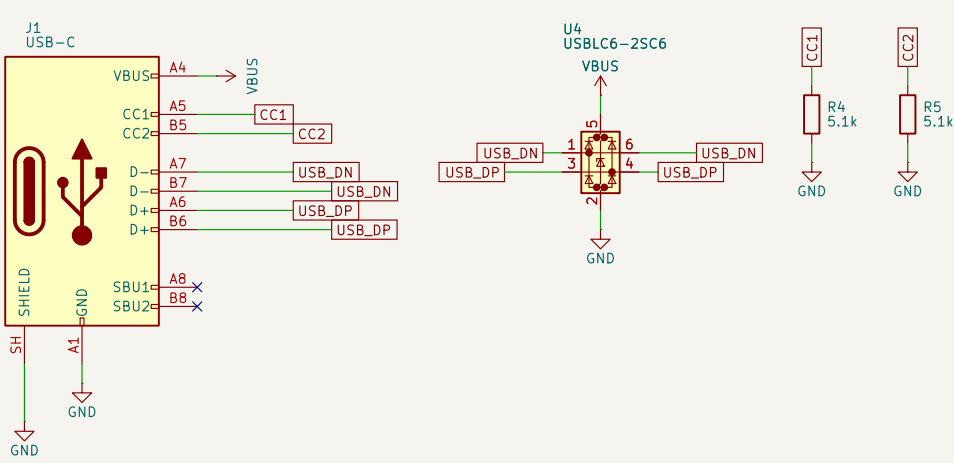


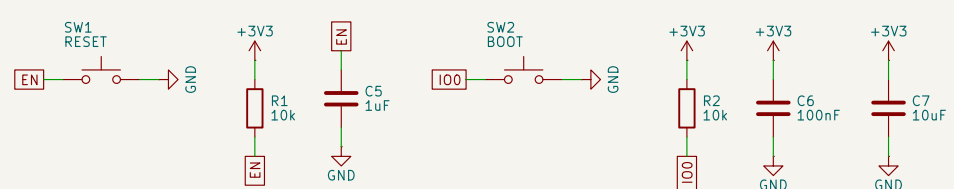
POWER IN
5V DC terminal or USB, Ored Schottky diodes



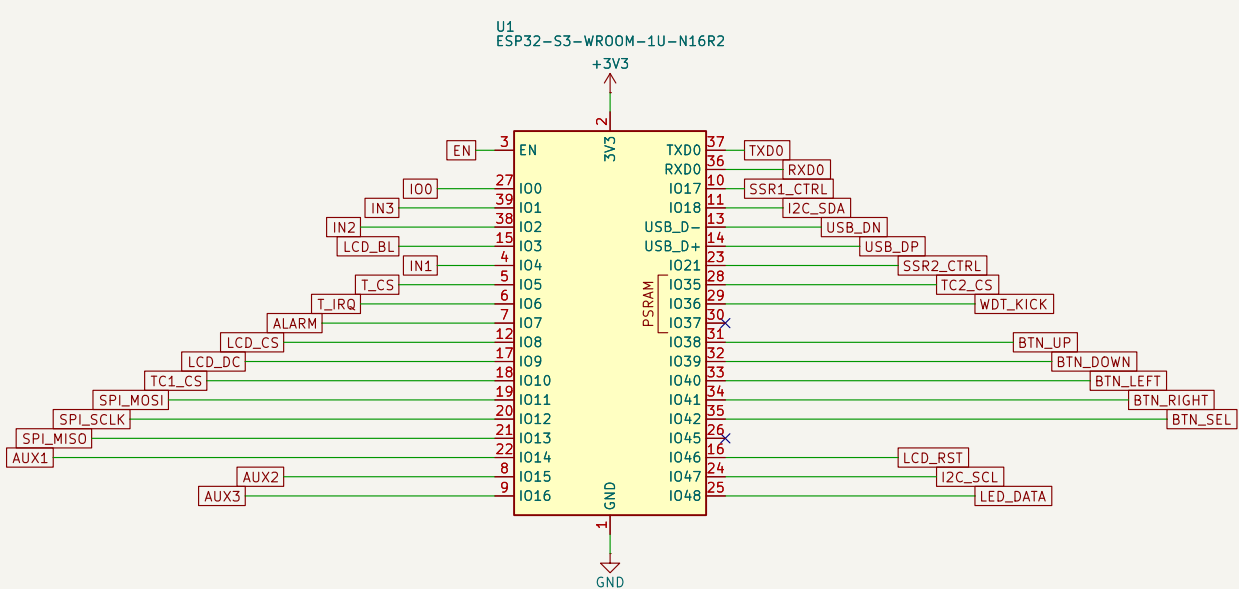
USB-C
native USB flashing + ESD



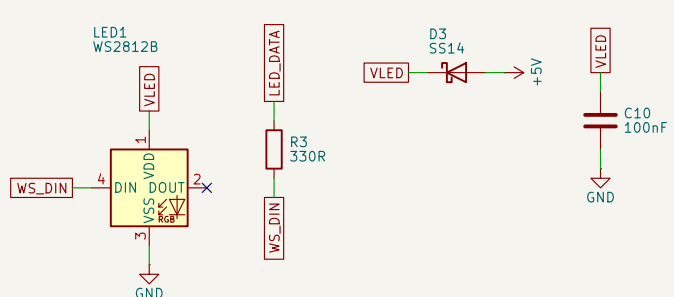
RESET / BOOT / DECOUPLING



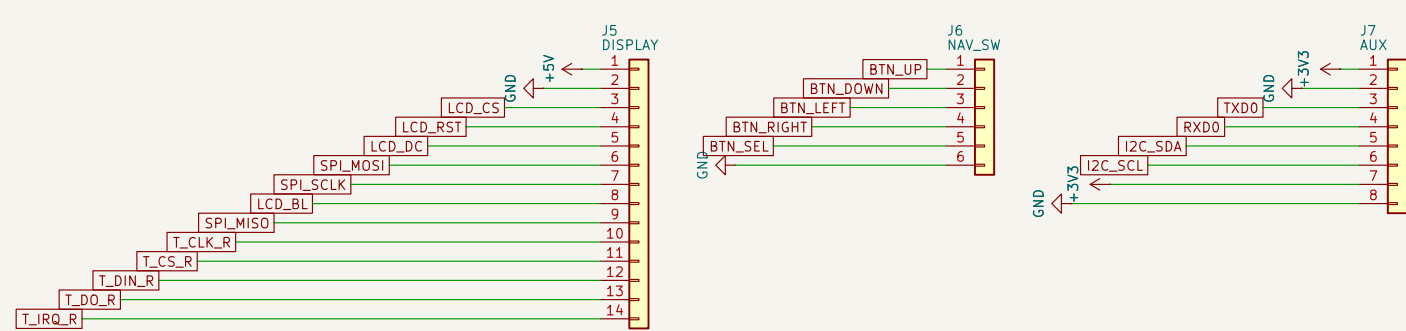
ESP32-S3-WROOM-1U-N16R2
GPIOs = firmware Kconfig defaults



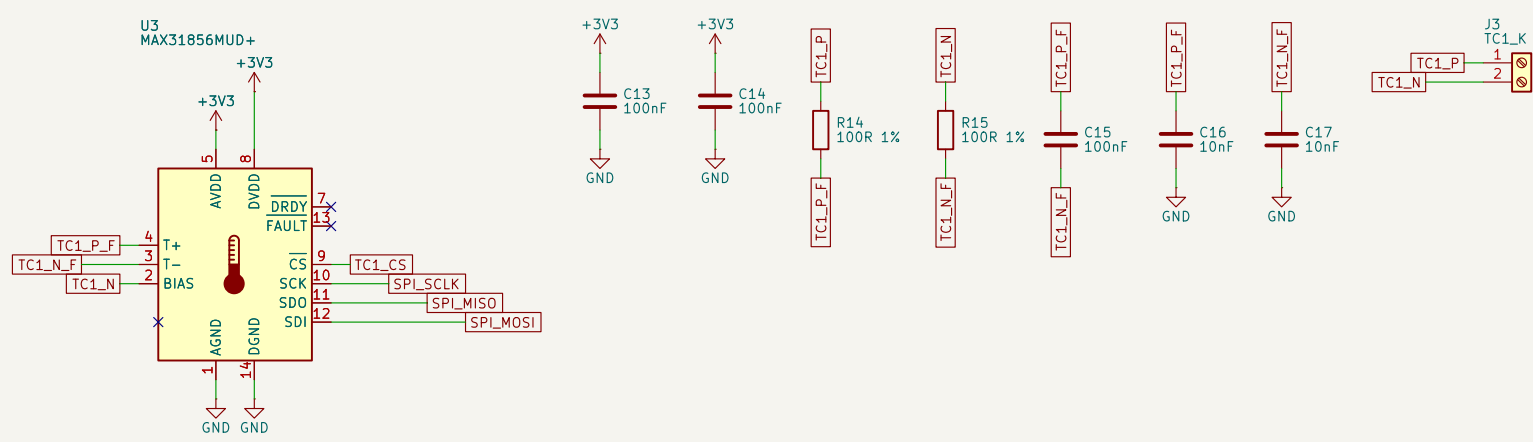
WS2812B STATUS LED
VDD dropped -4.6V for 3.3V data margin



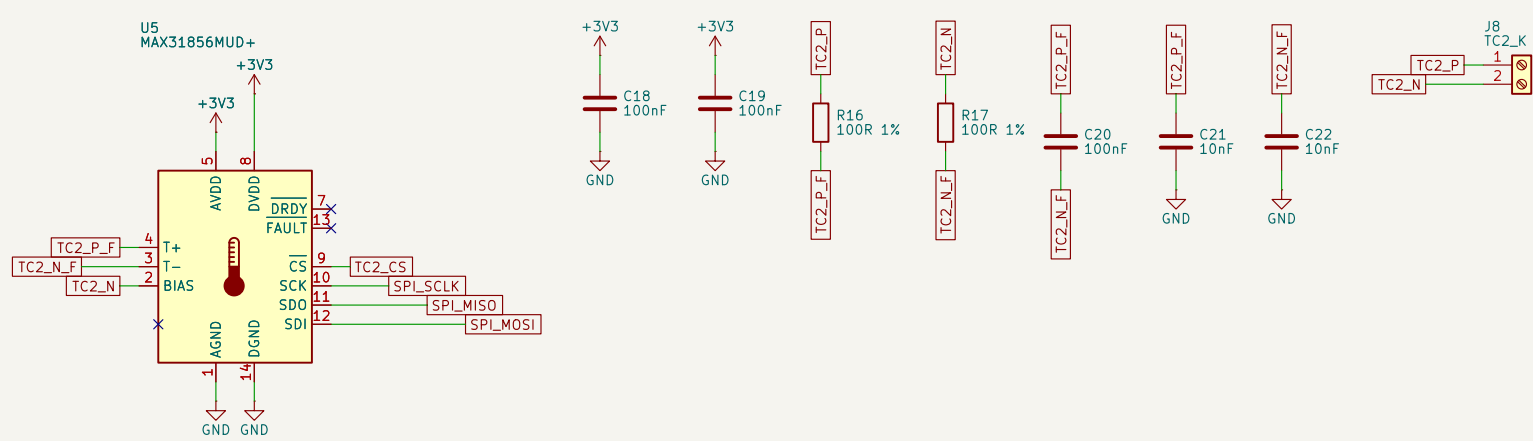
HEADERS
J5 display+touch (14-pin), J6 nav, J7 aux+I2C



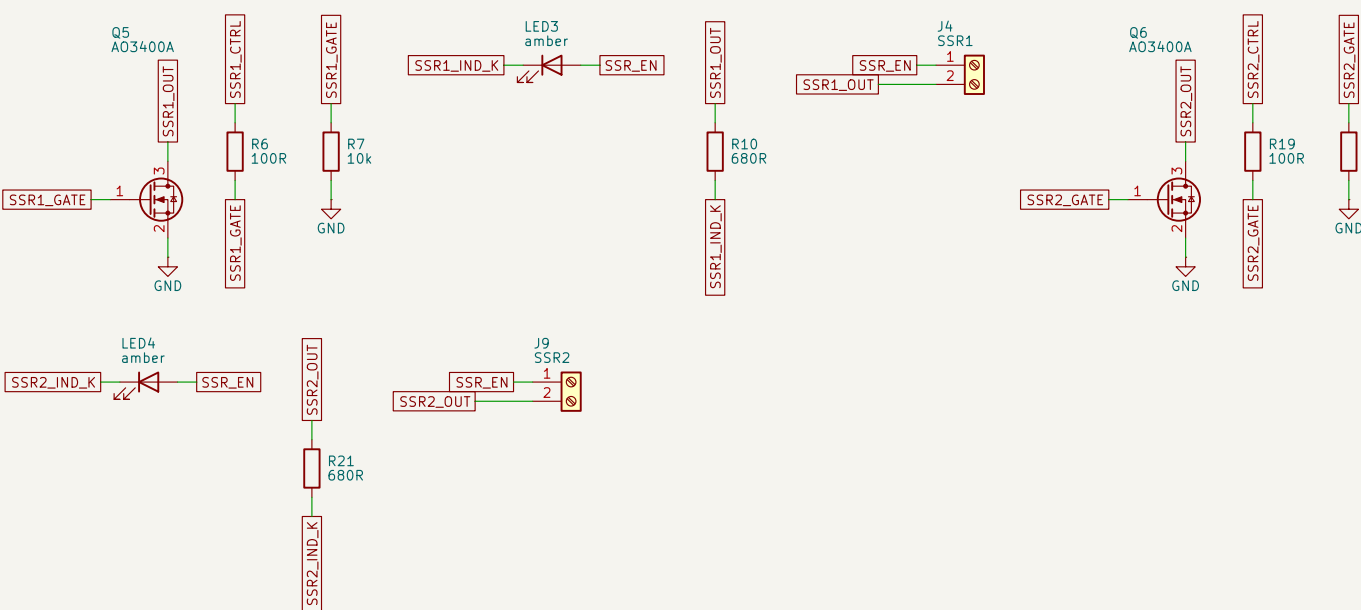
THERMOCOUPLE 1 (control) MAX31856
I- floats to J3, biased via BIAS



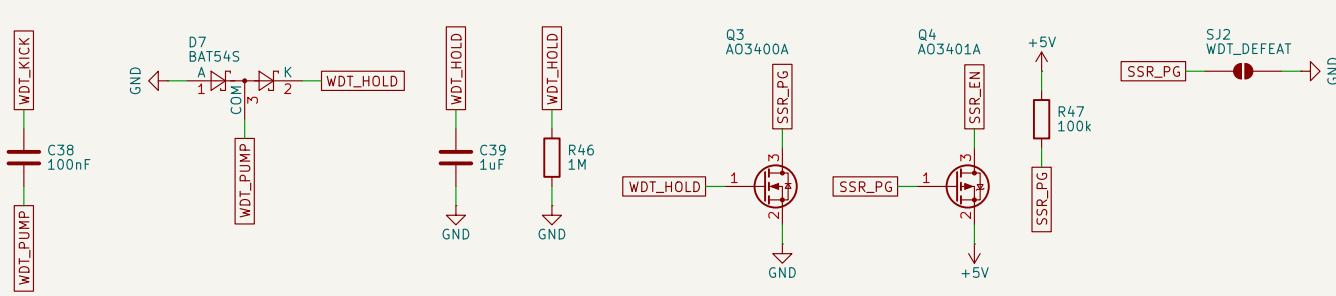
THERMOCOUPLE 2 (load) MAX31856
I- floats to J8, biased via BIAS



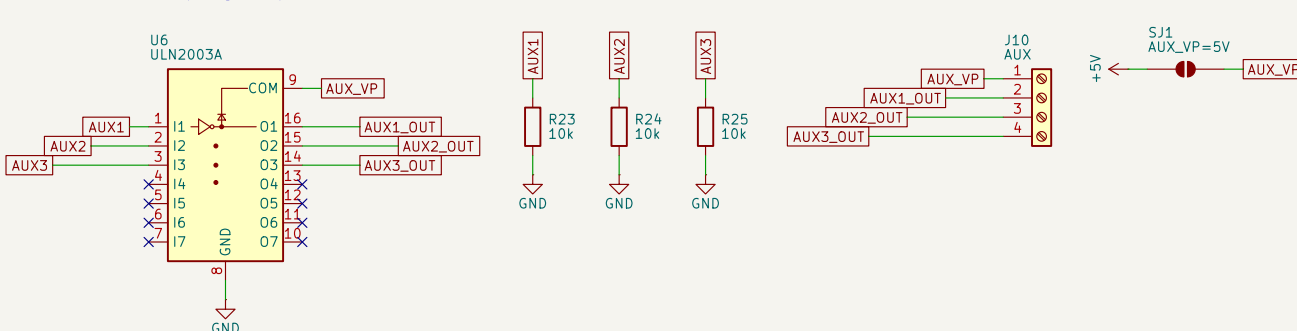
SSR DRIVE x2
low-side AO3400A; Indicator LED across each terminal



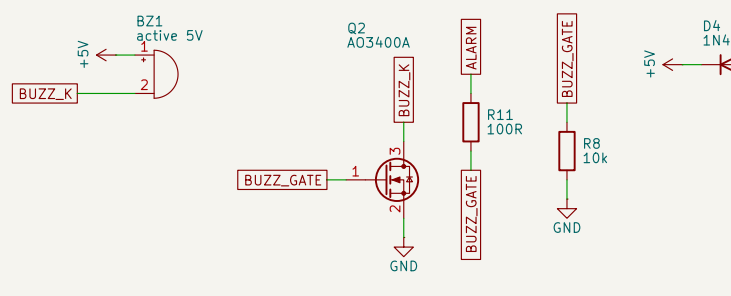
HARDWARE WATCHDOG
C38/D7/C39/R46 diode charge pump on WDT_KICK holds Q3 on; Q3 pulls SSR_P6 down, turning on high-side Q4, which supplies SSR_EN - the +5V rail feeding BOTH SSR terminals and both indicators. R47 is the fail-safe pull-up. S12 = bring-up defeat, REMOVE for service.



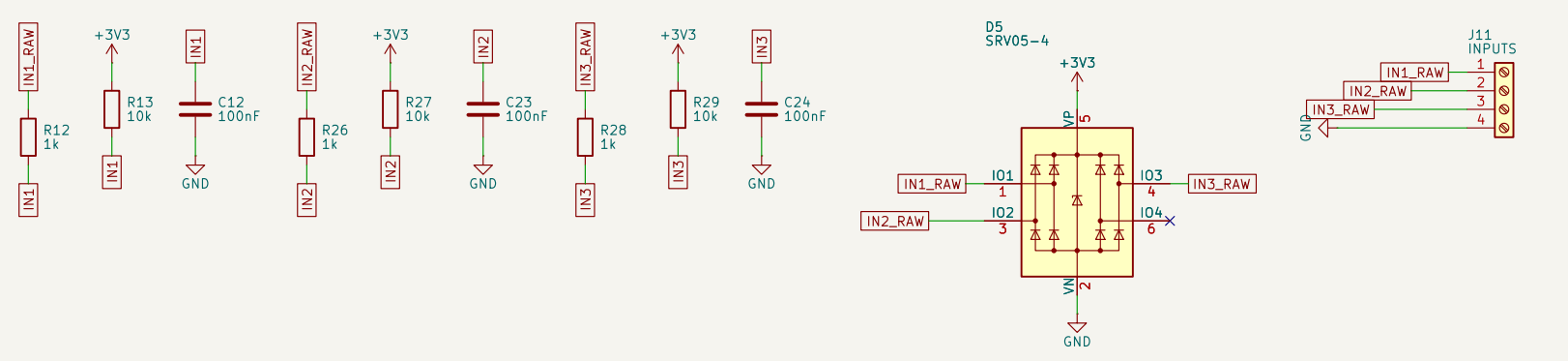
AUX OUTPUT BANK
ULN2003 vent/purge/spare; COM->AUX_VP, S11 links to +5V



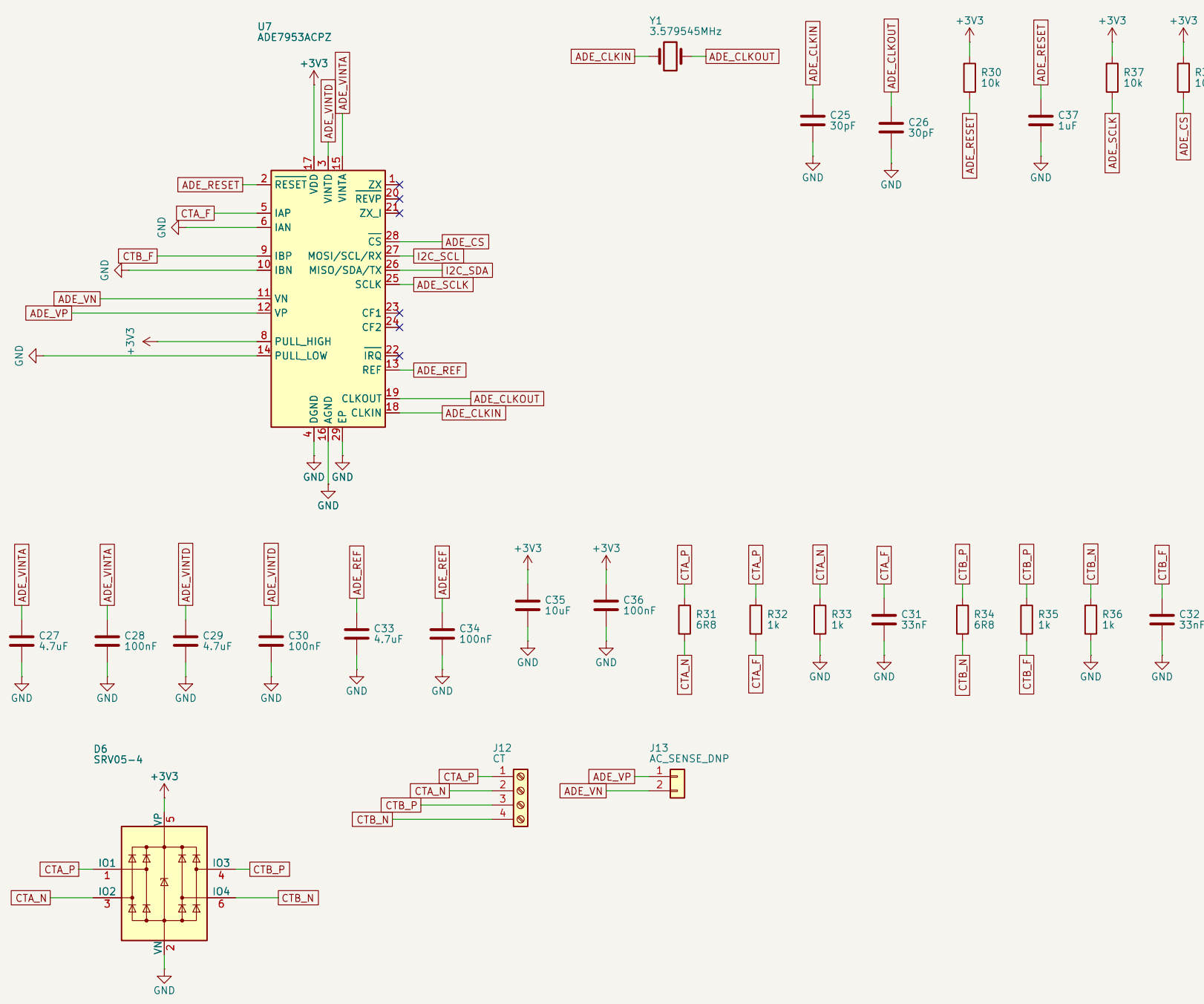
ALARM BUZZER



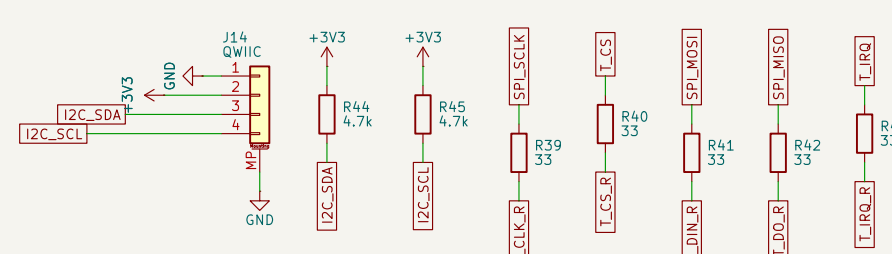
PROTECTED INPUTS x3 Iid / gas-flow / spare
1k + 10k pull-up + 100nF each, SRV05-4 TVS



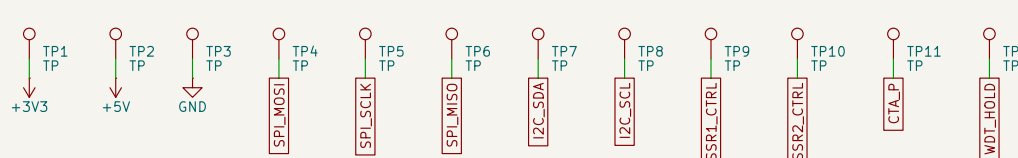
CT CURRENT SENSING ADE7953, I2C, current-only
no mains - VP/VN to DNP J13



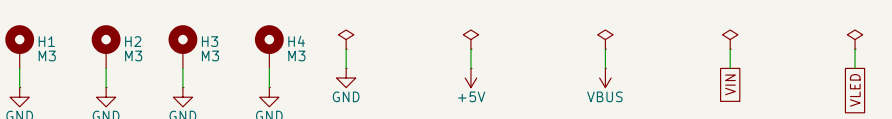
TOUCH DAMPING + I2C EXPANSION
R39-R43 damp the shared SPI2 bus for the display module's XPT2046 (not on this board); J14 Qwic and J7 5-8 (0.1 in) share the I2C bus, pulled up by R44/R45



TEST POINTS



MOUNTING / POWER FLAGS



NOTES - Bisque kiln controller, rev B

SSR terminals J4/J9; pin1 = SSR_EN (board +5V, watchdog-gated).
pin2 = the switched low side. Boot-safe: R7/R20 hold each MOSFET gate down while the ESP32 pins are high-impedance.
NOT opto-isolated - rev B tried that and reverted it: an opto only isolates if the SSR loop is powered off-board, and this board powers it.
TC1 terminal J3 / TC2 terminal J8: pin1 = K+, pin2 = K- - both float, biased near AGND only through each MAX31856's internal BIAS network. Ungrounded-junction probes required: two grounded-junction probes in one kiln would loop through it.
Nav switch J6 is panel-mounted: inputs use ESP32 pull-ups.
J11: IN1/IN2/IN3 (Iid/gas-flow/spare) + GND - dry contact each channel to GND.
Display J5 (14-pin, ST77965 + XPT2046 touch module, LCDWIKI MSP4021): 1=5V 2=GND 3=CS 4=RS1 5=DC 6=SDI/MOSI 7=SSCK 8=BL 9=SDO/MISO 10=I2C_CLK 11=I2C_CS 12=I2C_D0 13=I2C_D1 14=I2C_D2 15=I2C_D3 16=I2C_D4 17=I2C_D5 18=I2C_D6 19=I2C_D7 20=I2C_D8 21=I2C_D9 22=I2C_D10 23=I2C_D11 24=I2C_D12 25=I2C_D13 26=I2C_D14 27=I2C_D15 28=I2C_D16 29=I2C_D17 30=I2C_D18 31=I2C_D19 32=I2C_D20 33=I2C_D21 34=I2C_D22 35=I2C_D23 36=I2C_D24 37=I2C_D25 38=I2C_D26 39=I2C_D27 40=I2C_D28 41=I2C_D29 42=I2C_D30 43=I2C_D31 44=I2C_D32 45=I2C_D33 46=I2C_D34 47=I2C_D35 48=I2C_D36 49=I2C_D37 50=I2C_D38 51=I2C_D39 52=I2C_D40 53=I2C_D41 54=I2C_D42 55=I2C_D43 56=I2C_D44 57=I2C_D45 58=I2C_D46 59=I2C_D47 60=I2C_D48 61=I2C_D49 62=I2C_D50 63=I2C_D51 64=I2C_D52 65=I2C_D53 66=I2C_D54 67=I2C_D55 68=I2C_D56 69=I2C_D57 70=I2C_D58 71=I2C_D59 72=I2C_D60 73=I2C_D61 74=I2C_D62 75=I2C_D63 76=I2C_D64 77=I2C_D65 78=I2C_D66 79=I2C_D67 80=I2C_D68 81=I2C_D69 82=I2C_D70 83=I2C_D71 84=I2C_D72 85=I2C_D73 86=I2C_D74 87=I2C_D75 88=I2C_D76 89=I2C_D77 90=I2C_D78 91=I2C_D79 92=I2C_D80 93=I2C_D81 94=I2C_D82 95=I2C_D83 96=I2C_D84 97=I2C_D85 98=I2C_D86 99=I2C_D87 100=I2C_D88 101=I2C_D89 102=I2C_D90 103=I2C_D91 104=I2C_D92 105=I2C_D93 106=I2C_D94 107=I2C_D95 108=I2C_D96 109=I2C_D97 110=I2C_D98 111=I2C_D99 112=I2C_D100 113=I2C_D101 114=I2C_D102 115=I2C_D103 116=I2C_D104 117=I2C_D105 118=I2C_D106 119=I2C_D107 120=I2C_D108 121=I2C_D109 122=I2C_D110 123=I2C_D111 124=I2C_D112 125=I2C_D113 126=I2C_D114 127=I2C_D115 128=I2C_D116 129=I2C_D117 130=I2C_D118 131=I2C_D119 132=I2C_D120 133=I2C_D121 134=I2C_D122 135=I2C_D123 136=I2C_D124 137=I2C_D125 138=I2C_D126 139=I2C_D127 140=I2C_D128 141=I2C_D129 142=I2C_D130 143=I2C_D131 144=I2C_D132 145=I2C_D133 146=I2C_D134 147=I2C_D135 148=I2C_D136 149=I2C_D137 150=I2C_D138 151=I2C_D139 152=I2C_D140 153=I2C_D141 154=I2C_D142 155=I2C_D143 156=I2C_D144 157=I2C_D145 158=I2C_D146 159=I2C_D147 160=I2C_D148 161=I2C_D149 162=I2C_D150 163=I2C_D151 164=I2C_D152 165=I2C_D153 166=I2C_D154 167=I2C_D155 168=I2C_D156 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400=I2C_D388 401=I2C_D389 402=I2C_D390 403=I2C_D391 404=I2C_D392 405=I2C_D393 406=I2C_D394 407=I2C_D395 408=I2C_D396 409=I2C_D397 410=I2C_D398 411=I2C_D399 412=I2C_D400 413=I2C_D401 414=I2C_D402 415=I2C_D403 416=I2C_D404 417=I2C_D405 418=I2C_D406 419=I2C_D407 420=I2C_D408 421=I2C_D409 422=I2C_D410 423=I2C_D411 424=I2C_D412 425=I2C_D413 426=I2C_D414 427=I2C_D415 428=I2C_D416 429=I2C_D417 430=I2C_D418 431=I2C_D419 432=I2C_D420 433=I2C_D421 434=I2C_D422 435=I2C_D423 436=I2C_D424 437=I2C_D425 438=I2C_D426 439=I2C_D427 440=I2C_D428 441=I2C_D429 442=I2C_D430 443=I2C_D431 444=I2C_D432 445=I2C_D433 446=I2C_D434 447=I2C_D435 448=I2C_D436 449=I2C_D437 450=I2C_D438 451=I2C_D439 452=I2C_D440 453=I2C_D441 454=I2C_D442 455=I2C_D443 456=I2C_D444 457=I2C_D445 458=I2C_D446 459=I2C_D447 460=I2C_D448 461=I2C_D449 462=I2C_D450 463=I2C_D451 464=I2C_D452 465=I2C_D453 466=I2C_D454 467=I2C_D455 468=I2C_D456 469=I2C_D457 470=I2C_D458 471=I2C_D459 472=I2C_D460 473=I2C_D461 474=I2C_D462 475=I2C_D463 476=I2C_D464 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554=I2C_D542 555=I2C_D543 556=I2C_D544 557=I2C_D545 558=I2C_D546 559=I2C_D547 560=I2C_D548 561=I2C_D549 562=I2C_D550 563=I2C_D551 564=I2C_D552 565=I2C_D553 566=I2C_D554 567=I2C_D555 568=I2C_D556 569=I2C_D557 570=I2C_D558 571=I2C_D559 572=I2C_D560 573=I2C_D561 574=I2C_D562 575=I2C_D563 576=I2C_D564 577=I2C_D565 578=I2C_D566 579=I2C_D567 580=I2C_D568 581=I2C_D569 582=I2C_D570 583=I2C_D571 584=I2C_D572 585=I2C_D573 586=I2C_D574 587=I2C_D575 588=I2C_D576 589=I2C_D577 590=I2C_D578 591=I2C_D579 592=I2C_D580 593=I2C_D581 594=I2C_D582 595=I2C_D583 596=I2C_D584 597=I2C_D585 598=I2C_D586 599=I2C_D587 600=I2C_D588 601=I2C_D589 602=I2C_D590 603=I2C_D591 604=I2C_D592 605=I2C_D593 606=I2C_D594 607=I2C_D595 608=I2C_D596 609=I2C_D597 610=I2C_D598 611=I2C_D599 612=I2C_D600 613=I2C_D601 614=I2C_D602 615=I2C_D603 616=I2C_D604 617=I2C_D605 618=I2C_D606 619=I2C_D607 620=I2C_D608 621=I2C_D609 622=I2C_D610 623=I2C_D611 624=I2C_D612 625=I2C_D613 626=I2C_D614 627=I2C_D615 628=I2C_D616 629=I2C_D617 630=I2C_D618 631=I2C_D619 632=I2C_D620 633=I2C_D621 634=I2C_D622 635=I2C_D623 636=I2C_D624 637=I2C_D625 638=I2C_D626 639=I2C_D627 640=I2C_D628 641=I2C_D629 642=I2C_D630 643=I2C_D631 644=I2C_D632 645=I2C_D633 646=I2C_D634 647=I2C_D635 648=I2C_D636 649=I2C_D637 650=I2C_D638 651=I2C_D639 652=I2C_D640 653=I2C_D641 654=I2C_D642 655=I2C_D643 656=I2C_D644 657=I2C_D645 658=I2C_D646 659=I2C_D647 660=I2C_D648 661=I2C_D649 662=I2C_D650 663=I2C_D651 664=I2C_D652 665=I2C_D653 666=I2C_D654 667=I2C_D655 668=I2C_D656 669=I2C_D657 670=I2C_D658 671=I2C_D659 672=I2C_D660 673=I2C_D661 674=I2C_D662 675=I2C_D663 676=I2C_D664 677=I2C_D665 678=I2C_D666 679=I2C_D667 680=I2C_D668 681=I2C_D669 682=I2C_D670 683=I2C_D671 684=I2C_D672 685=I2C_D673 686=I2C_D674 687=I2C_D675 688=I2C_D676 689=I2C_D677 690=I2C_D678 691=I2C_D679 692=I2C_D680 693=I2C_D681 694=I2C_D682 695=I2C_D683 696=I2C_D684 697=I2C_D685 698=I2C_D686 699=I2C_D687 700=I2C_D688 701=I2C_D689 702=I2C_D690 703=I2C_D691 704=I2C_D692 705=I2C_D693 706=I2C_D694 707=I2C_D695 708=I2C_D696 709=I2C_D697 710=I2C_D698 711=I2C_D699 712=I2C_D700 713=I2C_D701 714=I2C_D702 715=I2C_D703 716=I2C_D704 717=I2C_D705 718=I2C_D706 719=I2C_D707 720=I2C_D708 721=I2C_D709 722=I2C_D710 723=I2C_D711 724=I2C_D712 725=I2C_D713 726=I2C_D714 727=I2C_D715 728=I2C_D716 729=I2C_D717 730=I2C_D718 731=I2C_D719 732=I2C_D720 733=I2C_D721 734=I2C_D722 735=I2C_D723 736=I2C_D724 737=I2C_D725 738=I2C_D726 739=I2C_D727 740=I2C_D728 741=I2C_D729 742=I2C_D730 743=I2C_D731 744=I2C_D732 745=I2C_D733 746=I2C_D734 747=I2C_D735 748=I2C_D736 749=I2C_D737 750=I2C_D738 751=I2C_D739 752=I2C_D740 753=I2C_D741 754=I2C_D742 755=I2C_D743 756=I2C_D744 757=I2C_D745 758=I2C_D746 759=I2C_D747 760=I2C_D748 761=I2C_D749 762=I2C_D750 763=I2C_D751 764=I2C_D752 765=I2C_D753 766=I2C_D754 767=I2C_D755 768=I2C_D756 769=I2C_D757 770=I2C_D758 771=I2C_D759 772=I2C_D760 773=I2C_D761 774=I2C_D762 775=I2C_D763 776=I2C_D764 777=I2C_D765 778=I2C_D766 779=I2C_D767 780=I2C_D768 781=I2C_D769 782=I2C_D770 783=I2C_D771 784=I2C_D772 785=I2C_D773 786=I2C_D774 787=I2C_D775 788=I2C_D776 789=I2C_D777 790=I2C_D778 791=I2C_D779 792=I2C_D780 793=I2C_D781 794=I2C_D782 795=I2C_D783 796=I2C_D784 797=I2C_D785 798=I2C_D786 799=I2C_D787 800=I2C_D788 801=I2C_D789 802=I2C_D790 803=I2C_D791 804=I2C_D792 805=I2C_D793 806=I2C_D794 807=I2C_D795 808=I2C_D796 809=I2C_D797 810=I2C_D798 811=I2C_D799 812=I2C_D800 813=I2C_D801 814=I2C_D802 815=I2C_D803 816=I2C_D804 817=I2C_D805 818=I2C_D806 819=I2C_D807 820=I2C_D808 821=I2C_D809 822=I2C_D810 823=I2C_D811 824=I2C_D812 825=I2C_D813 826=I2C_D814 827=I2C_D815 828=I2C